



HUSK-e® Terminal Truck

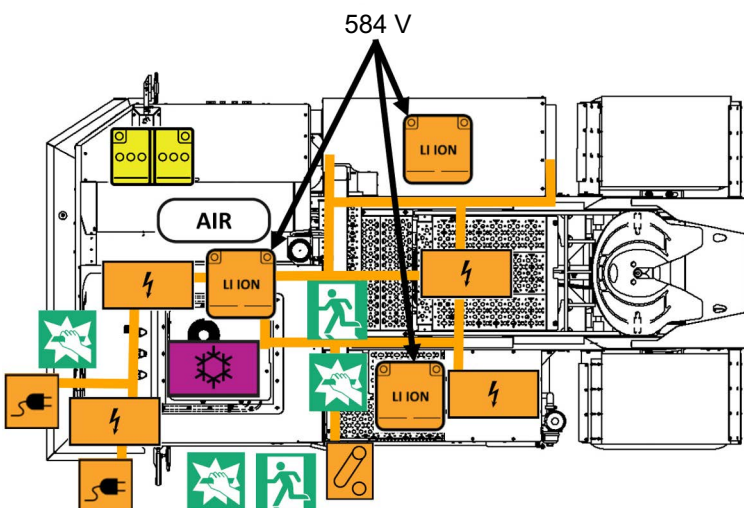
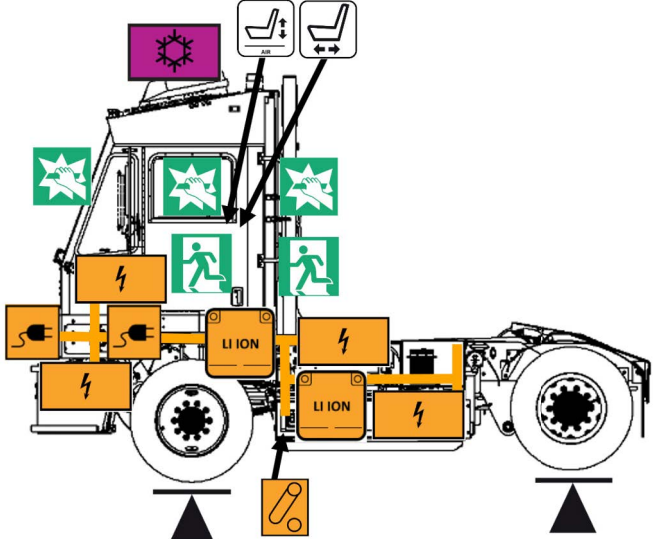
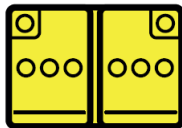


Emergency Response Guide

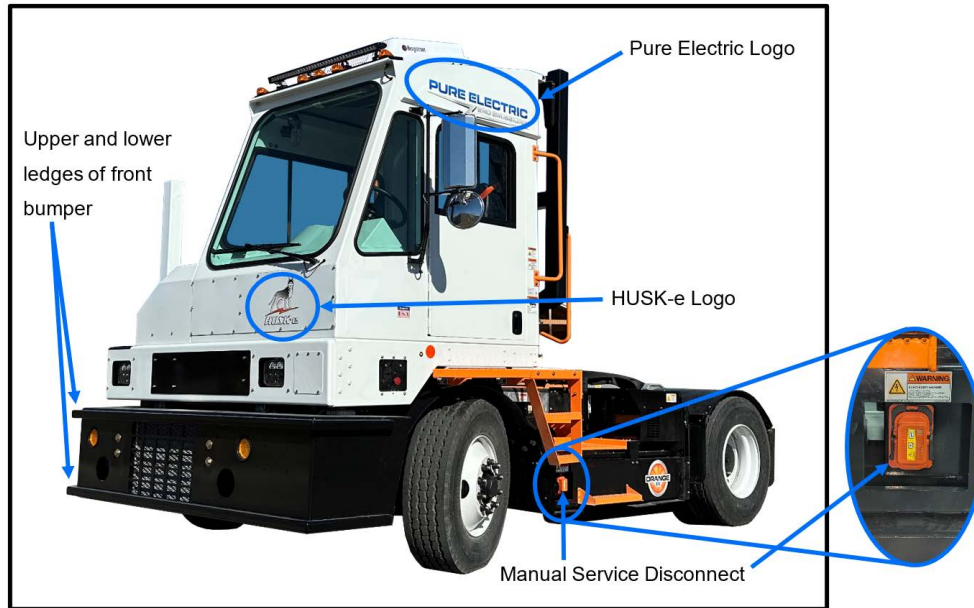


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0. Rescue Sheet

	Orange EV HUSK-e® Yard Truck Class 8 Truck Fully Electric 2024-Present				
					
 					
 Electric Propulsion	 High Voltage Li-Ion Battery	 High Voltage Component	 High Voltage Power Cable	 Disconnect High Voltage	 Low Voltage Battery
 Emergency Exit	 Break to Obtain Access	 Air Conditioning Component	 Lifting Point	 Seat Adjustment Height	 Seat Adjustment Longitudinal
 Charge Port	 Use Thermal Camera	 Compressed Air Tanks			
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1. Identification



Identification:

All HUSK-e® trucks have the following:

Unique bumper shape:

- All HUSK-e front bumpers are 22 inches tall and have an upper and lower ledge (2 inch square tube) wrapping around all three sides.
- The color of the front bumper may vary.

Manual Service Disconnect located below the driver's side ladder behind the front wheel.

Four warning stickers (two on the rear door and two on the lower right side when viewed from the back).

Vehicle Identification Number Plate showing the VIN and GVWR, located inside the cab, behind the driver side door.

Vehicle Emission Control Plate located inside the cab, above the driver side door.



Figure 1: HUSK-e Logo



Figure 2: Warning Stickers as Seen From the Back

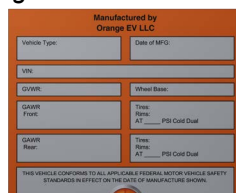


Figure 3: Vehicle Identification Number Plate



Figure 4: Vehicle Emission Control Information

2. Immobilization / stabilization / lifting

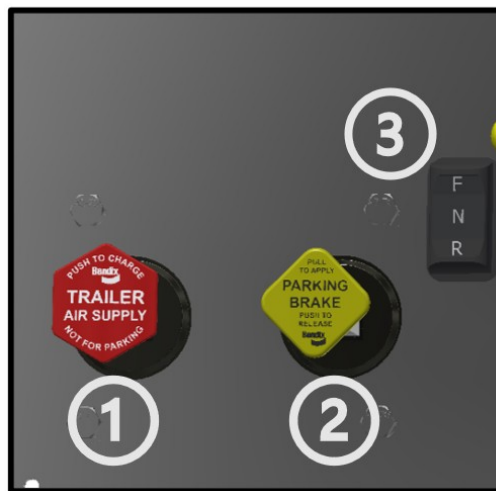
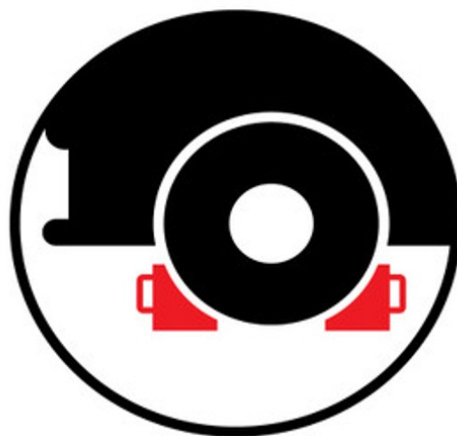


WARNING

Approach from the side to avoid truck travel path. A lack of noise does not indicate a lack of engine power. Silent movement remains possible until all immobilization steps are completed.

Immobilization:

1. Approach the truck from the side, chock in front and behind a wheel
2. Set trailer brake (1).
3. Set parking brake (2).
4. Shift truck into *N* (3), the middle position.



Lifting

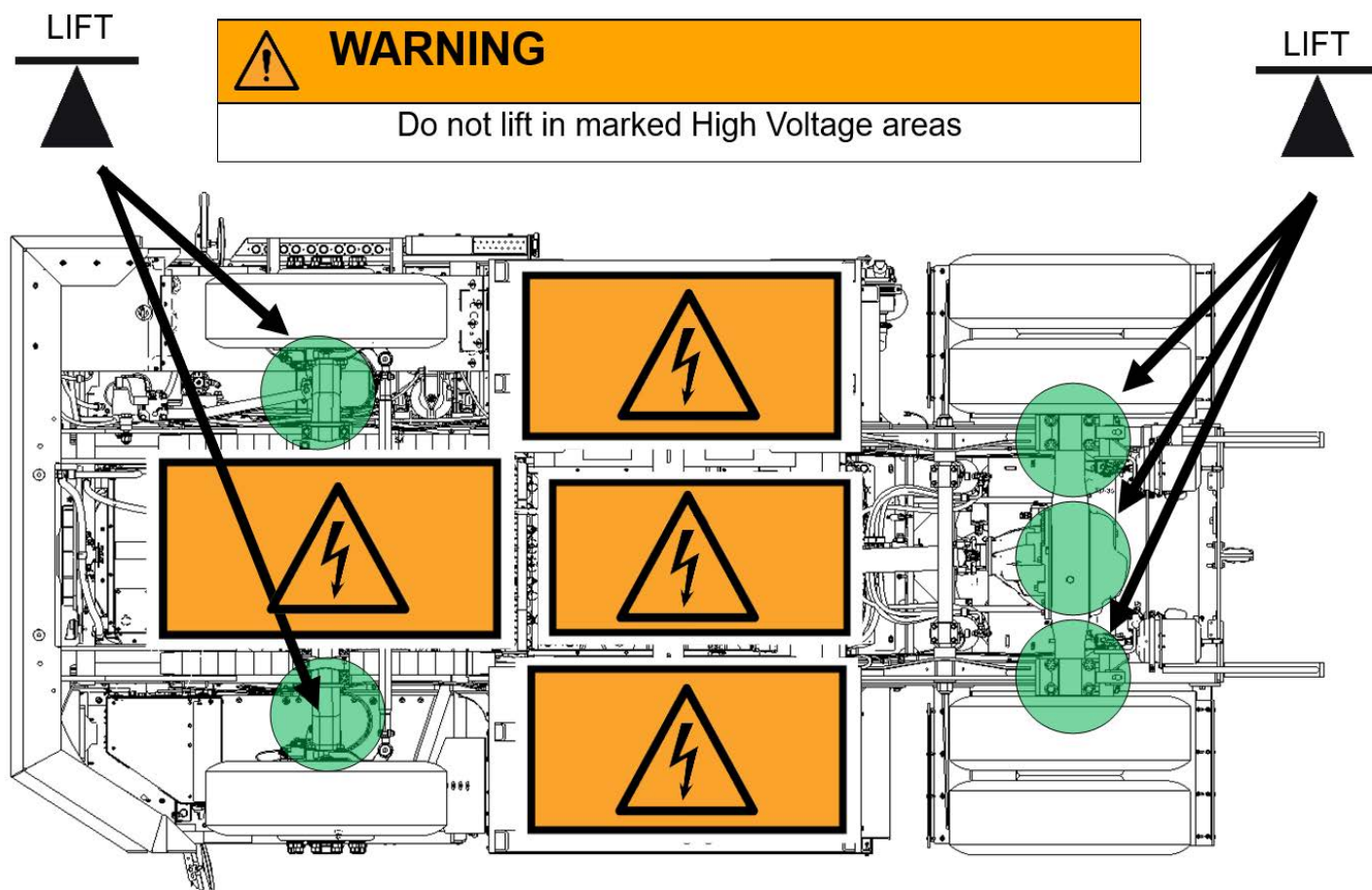


Figure 1: Bottom of Truck Shown

3. Disable direct hazards / safety regulations



WARNING

Assume the high voltage system is energized. Do not touch any high voltage cables or components. Approach from the side to avoid truck travel path. A lack of noise does not indicate a lack of engine power. Silent movement remains possible until all immobilization steps are completed.



Shutdown High Voltage at BOTH places noted below

In the event of fire, submersion, accident, or other emergency, shut down the high-voltage system by following the procedure below if it is safe to do so. To disable the High Voltage system all 5 of the following steps **must** be followed:

1. Disconnect from charger if connected.
2. Turn off ignition.
3. Remove key, if applicable.
4. Turn off 12 V battery disconnect.

NOTE: When in the off position (vertical), the 12 V disconnect can have a *Lockout Tagout* style lock or standard pad lock installed.



5. Remove the Manual Service Disconnect (MSD).
 - a. Pull out on the small orange clip in the middle of the black lever.
 - b. Pull the black lever down.
 - c. Pull the Orange MSD cover out.



4. Access to occupants

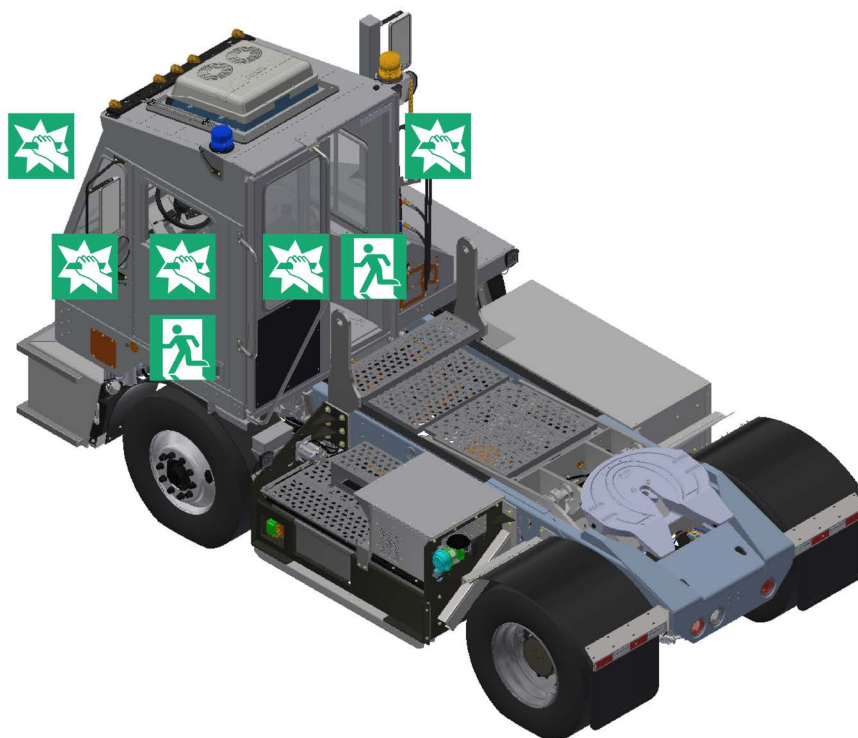


Figure 1: Two Door Exits: Driver's Side and Rear

Windows and Windshield

The windshield is made of laminated glass and can be kicked out to exit the cab in case of emergency.

The rear door is made of tempered safety glass.

The side glass, side door glass and fixed rear glass (behind the driver's seat) is tempered glass.

High Strength Zone

There is no high strength steel or Ultra High Strength Steel in the cab.

Seat Adjustment

The seat height is controlled using a switch on the front-left side of the seat. The fore and aft position is obtained by moving the seat cushion forward and back on its slides.

*Seat Height
Control*



DANGER

In the event of needing emergency access the cab via the rear door, ensure there is not a passenger in the training seat.

Training Seat (Optional)

If equipped, the training seat can be rotated forward for better access to the cab from the rear door.

To move the seat: Pull up on the entire seat assembly and rotate it forward.



Figure 2: Training Seat Rotated Out



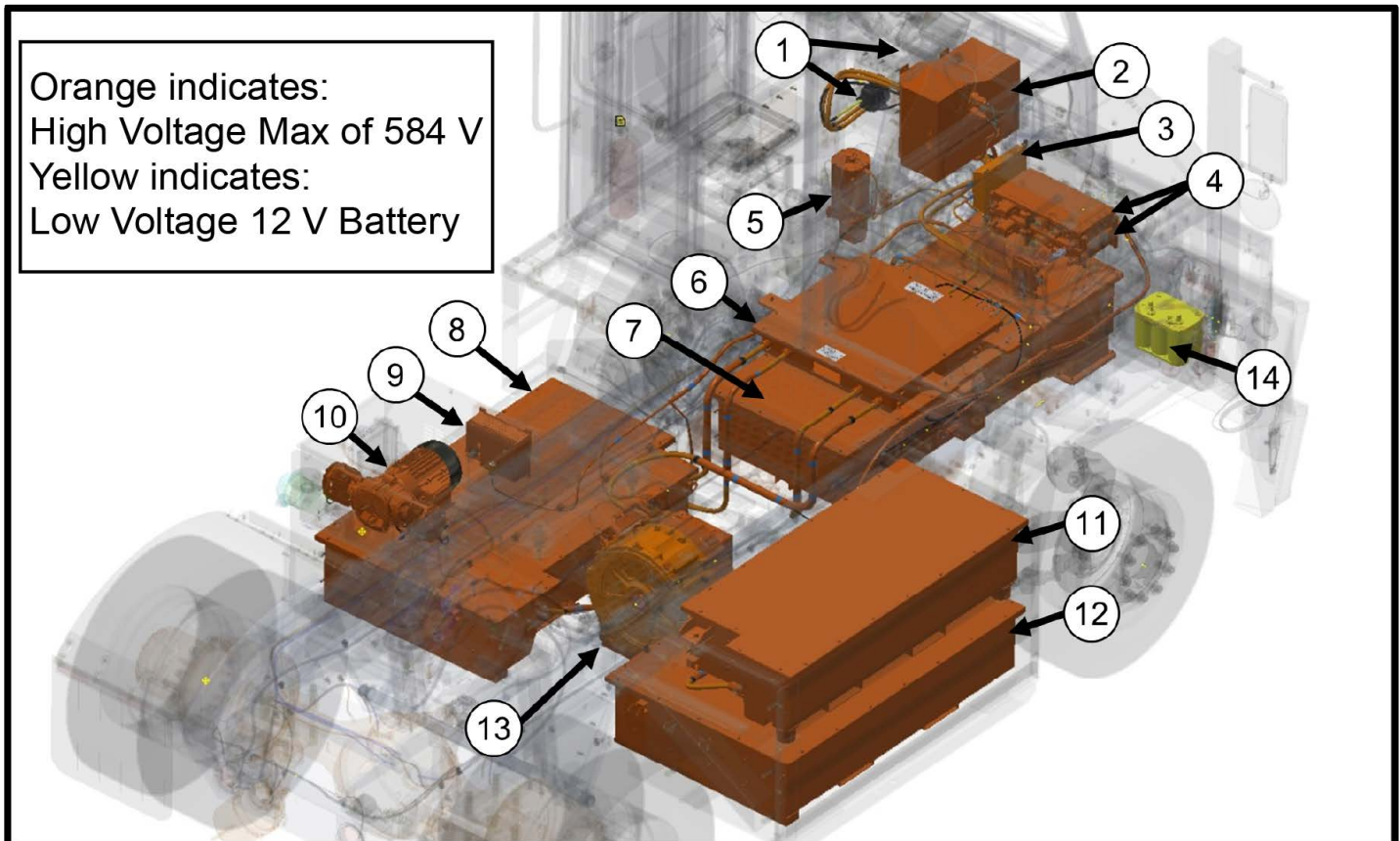
Figure 3: Training Seat Stowed (Rotated Forward)

5. Stored energy / liquids / gases / solids

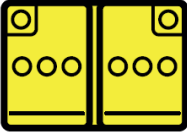


WARNING

The battery assembly covers should never be breached or removed under any circumstances, including fire. Doing so might result in severe electrical burns, shocks or electrocution.



1. Charging Receptacle, CCS1 Type (in front of driver's side corner of cab facing front or side.)
2. Cabin Heater
3. Hydraulic Pump Controller
4. DC to DC Converters (Quantity of 2 or 3)
5. Hydraulic Pump
6. Main Distribution Box
7. Front Battery Box
8. Driver Side Battery Box and Manual Service Disconnect (MSD located on outer front corner where arrow is pointing)
9. Air Compressor Controller
10. Air Compressor
11. Passenger Side Upper Battery Box
12. Passenger Side Lower Battery Box
13. Traction Motor Assembly
14. 12 V Battery (AGM Type)

Liquids / gases / solids					
Component Type	Type / Chemistry	Quantity	Volume / Weight	Maximum Operating Pressure	Specific Danger Symbol
584 V Traction Batteries 	Lithium-Ion	4 Battery Packs	Forward: 1122 lbs. Driver side: 1157 lbs. Passenger side upper: 833 lbs. Passenger side lower: 1157 lbs.		 
					 
12 V Battery 	AGM	1	43.5 lbs.		 
					 
Air System 	Compressed Air	3 Tanks	15.54 gal.	150 psi	
Coolant	Ethylene Glycol 60% / Water 40%		6 gal.	5 psi	
A/C Refrigerant 	R-134a		1100 grams	300 psi	

Flammable Liquids					
Component Type	Type / Chemistry	Quantity	Volume / Weight	Maximum Operating Pressure	Specific Danger Symbol
A) Hydraulic Fluid	Dexron 3 or Dexron 6 Automatic Transmission Fluid	1 Tank	18.5 gal.		 
B) Rear Axle	API GL-5	1 for single rear axle truck 3 for tandem axle truck	XP (Port) 5.76 gal. Rail 6.45 gal. Tandem 6.08 gal.		 
C) Interior Materials	In accordance with DOT/FMVSS 302				

	Structural PPE does not offer electrical protection. Never open high voltage components or cut orange high voltage cables.
	The presence of a sweet, ether-like odor could be due to a leaking lithium-ion battery. Liquid on the ground around the truck is not an indicator of a battery leak. Lithium-ion electrolyte will form droplets, not puddles.
	Damage to the pressurized hydraulic fluid system, or compressed air system could create high pressure fluid leaks and flying debris.

6. In case of fire



Use large, sustained quantities of water to extinguish any vehicle fire and cool down battery enclosures.



Use a thermal imaging camera to observe high voltage components. Ensure that the battery box has cooled sufficiently and is not increasing in temperature, indicating potential thermal runaway of the lithium-ion batteries. Monitor for hissing, popping, or smoke.



Avoid direct contact with high voltage battery assembly, cables, and components.

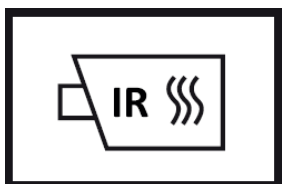


Batteries can release toxic fumes under fire conditions. Wear appropriate inhalation protection.



Special Battery Access

Open hood on outside of passenger side of cab to get water to battery or high voltage components.



Lithium-ion batteries contain liquid, flammable electrolyte.

Burning batteries can also ignite other batteries in the vicinity.

The extinguishing agent must be applied continuously until fully cooled down; otherwise, there will be a risk of a new ignition.

- A burning lithium-ion battery generally cannot be extinguished directly. Water as the extinguishing agent can be used for cooling lithium-ion batteries.
- A battery fire may continue to burn for several hours or re-ignite, so it is recommended to continue to cool the battery with excessive amounts of water.
- The temperature of the battery can be monitored with an thermal imaging camera to ensure it is not heating up.
- Re-check the temperature of the battery once an hour to ensure the temperature is not increasing. If the temperature is increasing continue to cool the battery with water.

Do not store a vehicle containing a damaged or burning lithium-ion battery within 15 feet of a structure or other vehicles.

7. In case of submersion



DANGER

Handling a submerged vehicle without appropriate training and personal protective equipment (PPE) can result in serious injury or death due to electric shock.

- Submersion in water can damage 12 V, 24 V and 584 V components.
- The damage level of a submerged vehicle may not be visible.
- Avoid any contact with the traction voltage cables and electric components.
- If possible, disable direct hazards (See Section 3).



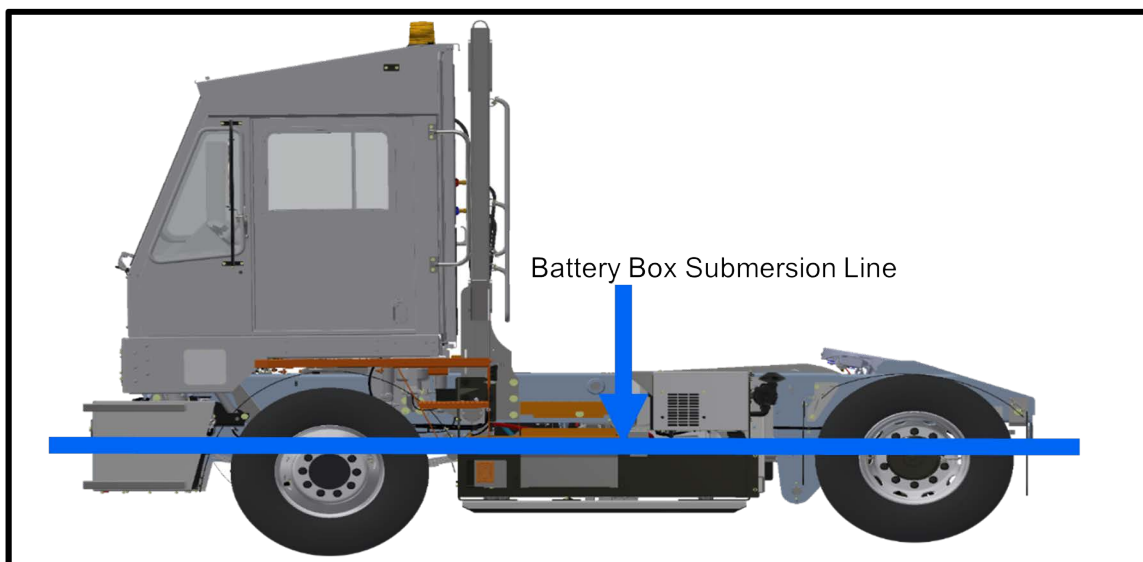
WARNING

Only enter the water if properly equipped with PPE. Avoid contact with high voltage battery assembly, cables, and components.



WARNING

Submersion, especially in salt water, can damage components and create a fire risk.



8. Towing / transportation / storage



WARNING

Avoid contact with high voltage components and cables. In the case of a fire, relays could have fused closed due to heat, keeping the truck energized even after disabling.



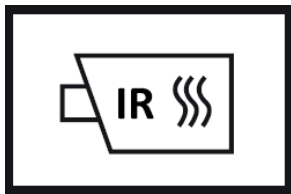
WARNING

The battery assembly covers should never be breached or removed under any circumstances, including fire. Doing so might result in severe electrical burns, shocks or electrocution.



WARNING

If the traction batteries are damaged, there is a risk of thermal or chemical reaction.



- Park the fully electric truck involved in an accident in a suitable place maintaining a safe distance from other vehicles, buildings and combustible objects.
- Risk of delayed fire can happen after the fire suppression or if the lithium-ion batteries are damaged.
- Observe the truck for a minimum period of 48 hours using a thermal infrared camera.

Transporting the Truck

Transporting a truck requires special equipment and training. Orange EV recommends that a professional transport service (or trained service professionals) be used to transport a damaged truck.

While the non-operating truck may be moved onsite a short distance with all six wheels on the ground, for travel off site, Orange EV recommends transporting the truck with all wheels off the ground.



WARNING

The electric motor can produce electricity when moving the truck with the rear drive tires on the ground.

This remains a potential source of electric shock even when the high voltage system is disabled.

Before towing the truck, it is mandatory to disconnect the drive to the wheels.

The drive to the wheels is disabled by uncoupling the drive shaft from the rear axle by removing the driveshaft, see ***Removing the Driveshaft*** below.

Towing truck with rear wheels suspended

If it is not possible to transport the truck with all six wheels off the ground, the secondary option is to tow the truck with the rear wheels suspended. The steering axle must be locked in the straight-ahead position. If applicable, spring brakes on the front axle must be caged. To cage the brakes, see ***Caging the Brakes*** below.

Towing truck with front wheels suspended

The driveshaft must be removed when towing with the front wheels suspended. To remove it, see ***Removing the Driveshaft*** below.



WARNING

Obey the following when towing an Orange EV truck with the front wheels lifted and the rear wheels on the ground.

1. Use a rigid towing bar or properly restrain the towed truck. Using a chain or cable to tow the truck is not recommended.
2. Remove the driveshaft (see ***Removing the Driveshaft*** below).
3. Cage the spring brakes on all rear drive wheels. (See ***Caging the Brakes*** below). If the brakes are not caged, then a loss of air pressure while towing would apply the spring/ parking brakes and lock the rear wheels.

Removing the Driveshaft

To remove the driveshaft:

1. Locate the driveshaft in front of the rear axle. Driveshaft bolt locations are shown below.
2. Remove the bolts at the rear of the driveshaft where it is bolted to the rear axle.
3. Remove the bolts at the front of the driveshaft where it is bolted to the powertrain.

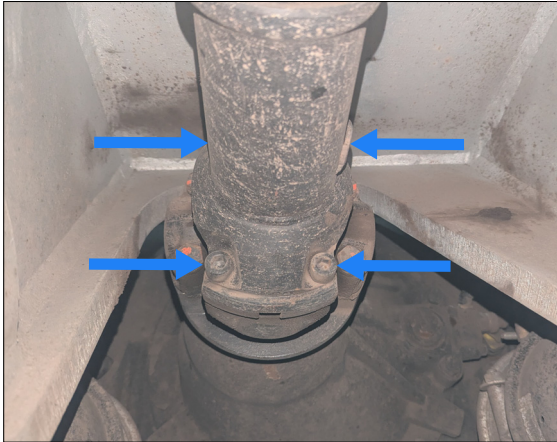


Figure 1: Bolts at Rear of Driveshaft

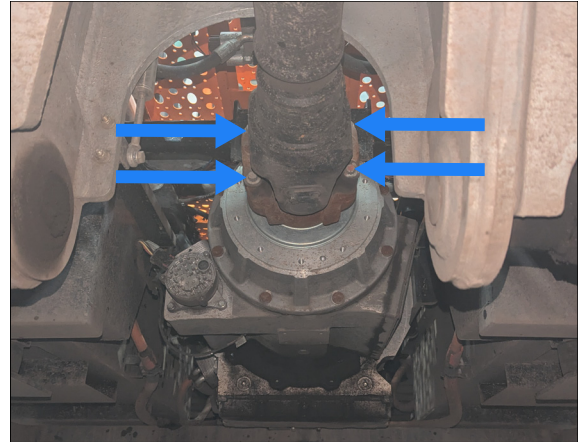


Figure 2: Bolts at Front of Driveshaft

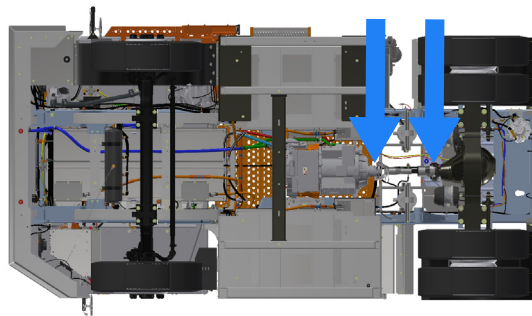


Figure 3: Driveshaft Bolt Locations as Seen From Below the Truck

Caging the Brakes

Integrated Caging Bolt

1. Locate the bolt on each brake chamber of the axle. There are two brake chambers on each axle.
2. Back out the bolt until it stops (about 76 mm / 3 inches), compressing the internal spring.

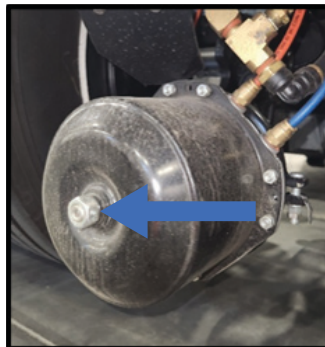


Figure 4: Integrated Caging Bolt on Internally Caged Brakes

External (removable) caging tool:

1. Locate the caging tool, typically mounted to the side of the brake chamber and remove it from its storage location (Figure 5).



Figure 5: External Caging Tool and Key Hole

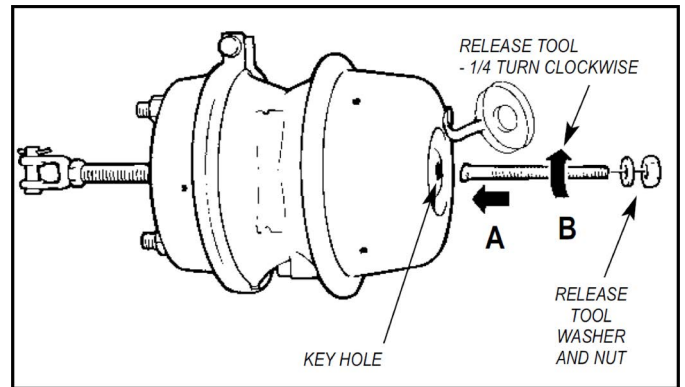


Figure 6: External Caging Tool Installation

2. Remove the access plug from the end of the brake canister.
3. Insert the "T" shaped end of the caging tool into the key hole. (A in Figure 5).
4. Turn the caging tool a quarter turn to engage with the slot on the pressure plate.
5. Try to pull the caging tool out; it should stay in. If it pulls out, repeat steps 3 and 4.
6. Thread the nut and washer down onto the threaded end of the caging tool all the way to the canister.
7. Tighten the nut all the way, the threaded part of the release tool should extend approximately 3" to 4" past the nut.

Storing the Truck

Park the damaged truck a safe distance from combustible objects, buildings, and other vehicles until the battery can be properly discharged. Notify the tow operator of stowing distance.



A lithium-ion battery on a severely damaged truck can increase in temperature over time and pose a delayed fire risk. Monitor battery temperature for 48 hours after incident using a thermal infrared camera.

9. Important additional information

Electric Shock



Never cut an orange cable.

Never cut a black cable with orange loom.

Do not perform any operation on a damaged truck without appropriate Personal Protective Equipment (PPE).

Call OEV Service at (866) 688-5223 for assessment of truck post incident or for guidance about storing a damaged truck.

Roll Over Protection (ROPS)

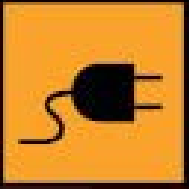
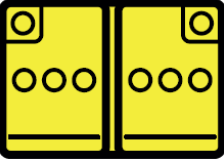
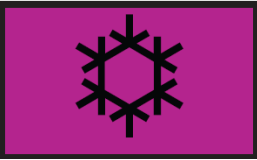


Some HUSK-e trucks are optionally equipped with Roll Over Protection Systems (ROPS). Trucks with the ROPS option have the large bar behind the cab as shown.

Fall On Protection (FOPS)

All HUSK-e trucks are equipped with Fall On Protection Systems (FOPS) compliant cabs.

10. Explanation of pictograms used

	Electric Propulsion		Left Hand Drive
	High Voltage Disconnect		Charge Port
	High Voltage Lithium Ion Battery		Low Voltage Battery
	High Voltage Component		High Voltage Power Cable
	Compressed Air Tank		Air Conditioner
	Lift Point		Emergency Exit
	Break to Obtain Access		Seat Adjustment Air System

	Seat Adjustment Longitudinal		Use Thermal Camera
	Warning Electricity		Flammable
	Acute Toxicity		Corrosives
	Explosive		Hazardous to Human Health
	Gases Under Pressure		Use Water to Extinguish Fire
	Environmental Hazard		Special Battery Access